

Debugger

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Debugger

An application to debug another application

How does it work?

- In Java the JVM has a debugging API that allows debugger tools to communicate with the application
- The debugger attaches to your application process on the JVM
- The debugging API allows the debugger to add breakpoints to the execution and read the memory context of the running program
- Mapping between the bytecode and the actual code written by you is made with the help of symbol files
- You need to compile your app in debug mode so that symbol files are generated.

Debuggers

Types

- Local and Remote
- Command line and Graphical interface
- Independent Application or Tool inside the IDE

Debugger

Capabilities:

- Suspend the target application at a given statement (using breakpoints)
- Run the target application instruction by instruction
- Inspect and change variables at runtime
- Examine the call stack at a given point of execution
- Evaluate expressions on the fly (using variables and methods of the target application)
- Inspect context of target application (threads, allocations, used memory, etc...)

Debugger

Downsides:

- Slows the performance of the target application
- Usually only for development environment
- Only doable for developers
- Can be a way of wasting a lot of time if not used with the right mindset

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